

Every Coin Counts

Counting coins on a textured pad with classical image processing

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01 Introduction

People count coins at a glance; a computer sees only pixels. These phone photos add three traps: a white pad full of folds, dark or motion-blurred shots, and round buttons the size of a coin.

Goal count only the coins with fixed classical rules, with no training and no machine learning.

Question can such rules cope with these conditions, and which objects does the program get wrong?

02 Method

- **Background:** a very large median filter models the pad and light; each pixel gets a Lab difference score.
- **Blobs:** multi-scale Laplacian of Gaussian finds round, coin-sized peaks.
- **Folds:** a Hessian shape test rejects ridges; grayscale opening removes seams.
- **Touching coins:** Hough circles separate what LoG merges.
- **Not a coin:** black-hat finds sewing holes; blue or pink is rejected.
- **Evaluation:** 26 photos, 328 hand-counted coins; every circle checked as TP, FP or FN.

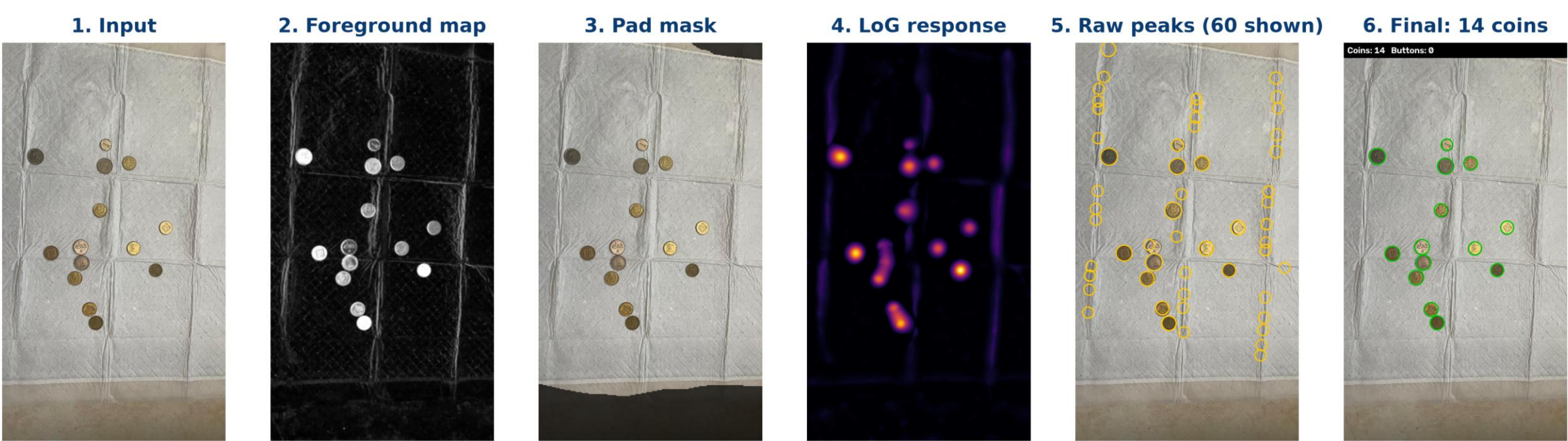
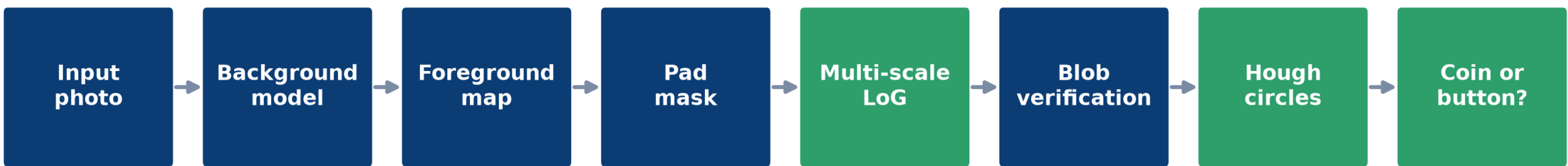
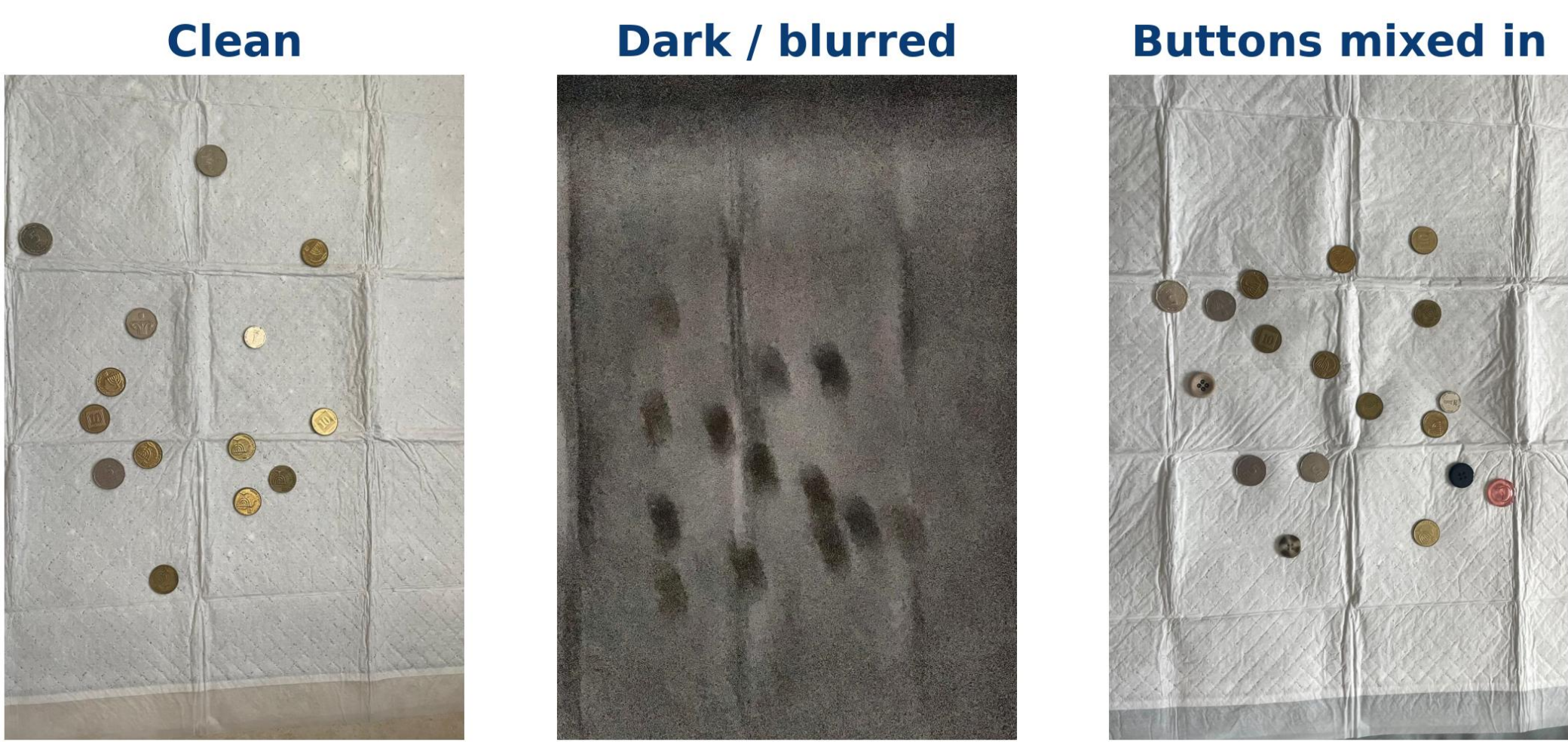


Photo → foreground map → pad mask → LoG → raw peaks (mostly folds) → 14 coins

03 Results

89.6%

recall

294 of 328 coins found

88.3%

precision

294 of 333 detections

39 / 34

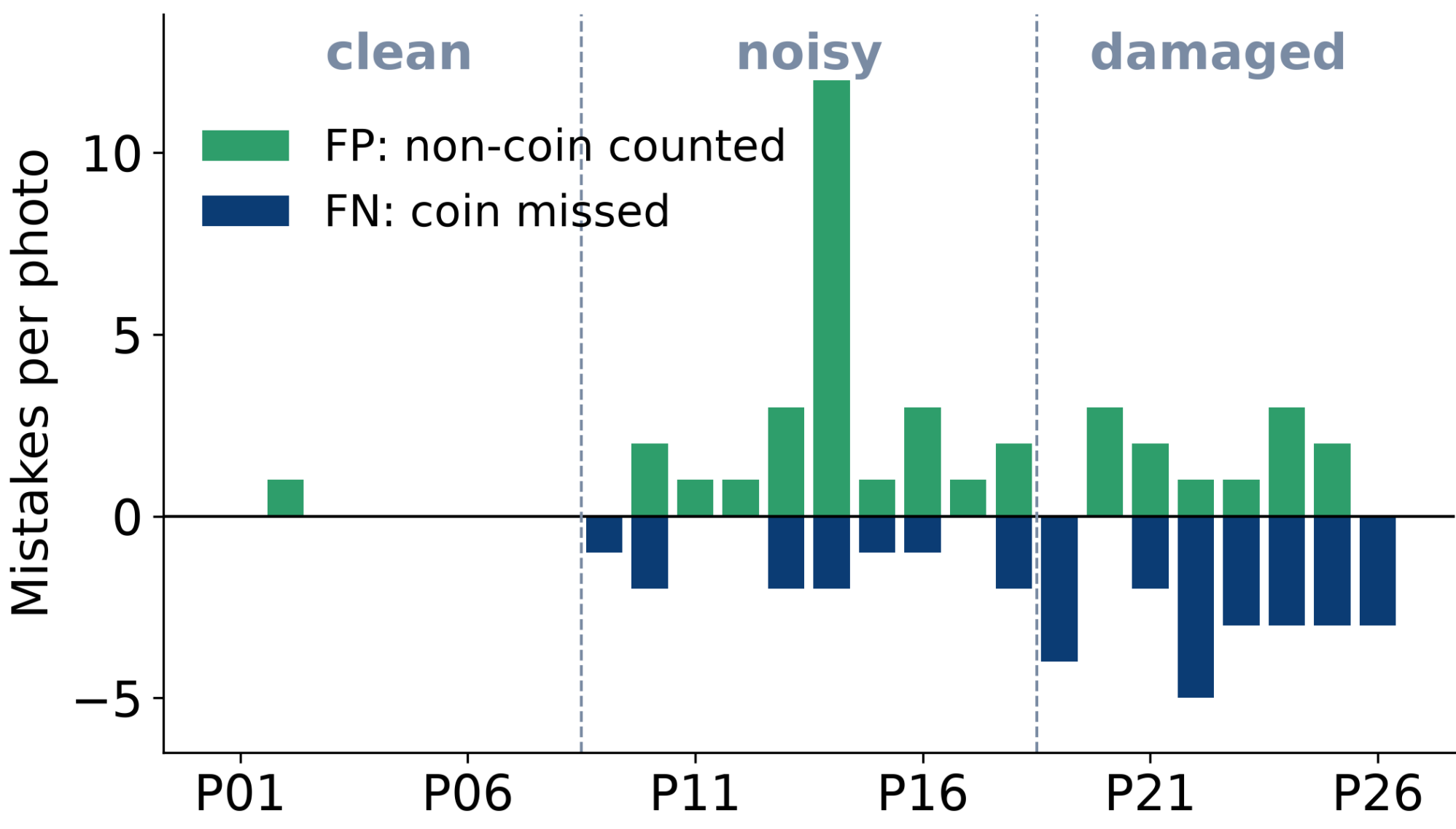
false / missed

FP non-coins · FN coins

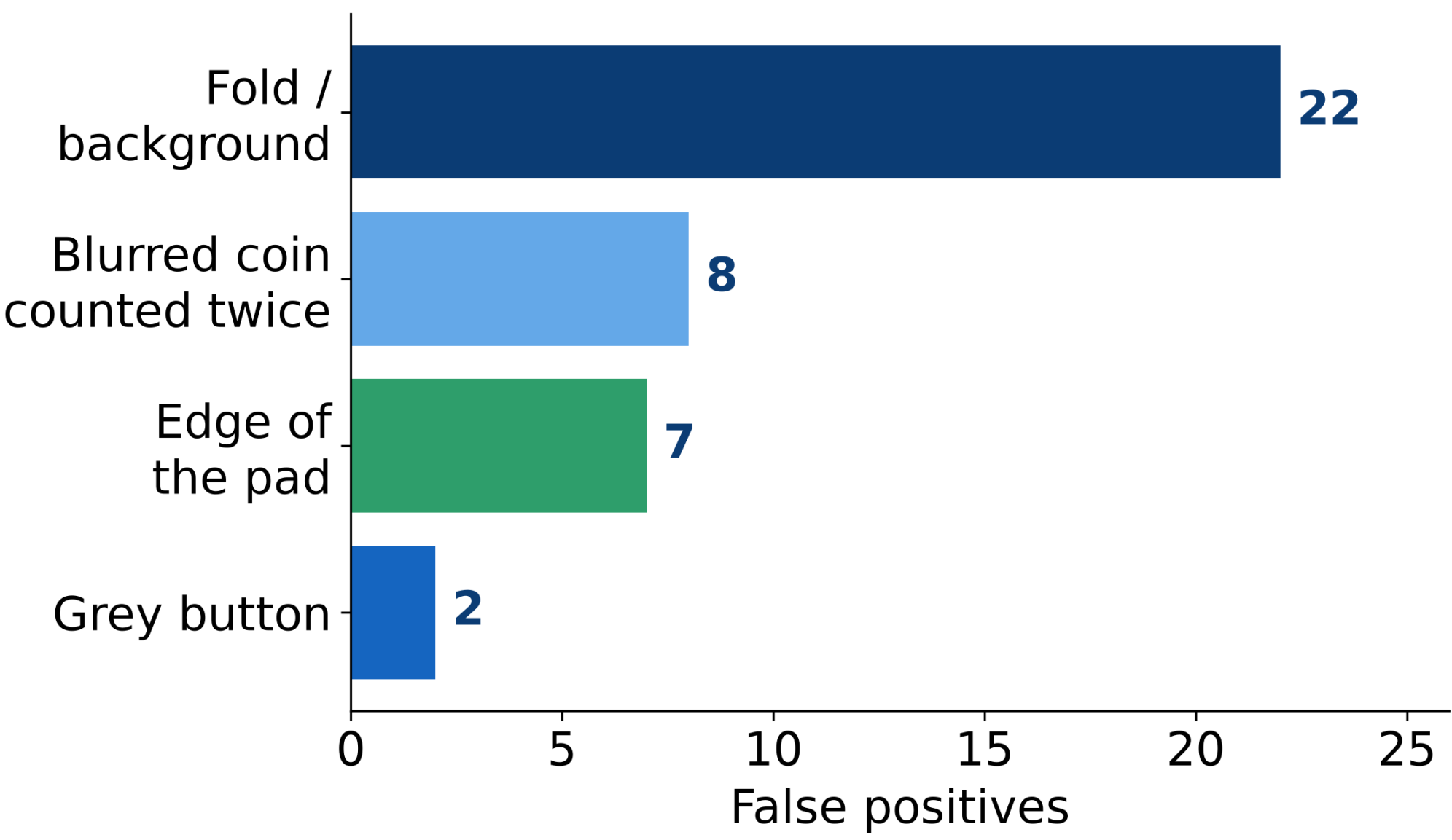
12/26

exact counts

only 7 with no mistake



False detections (up) and missed coins (down) per photo



What was mistaken for a coin (39 FP)

04 Conclusions

Target: count only the coins under changing conditions.

Achieved: 89.6% recall, 88.3% precision with classical rules only; all 98 coins found on clean photos and 9 of 12 buttons rejected. Dark photos add false coins; piles hide real ones.

05 Discussions

Next: merge detections along the blur direction, drop the pad's white hem from the mask, fit partial circles at the frame, and use rim colour for grey buttons.



Source code